

NOVOKUZNETSK, Russian Federation

WMO: 298460

Lat: 53.817N

Lon: 86.883E

Elev: 1010

StdP: 14.17

Time Zone: 7.00 (E07)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-23.1	-32.8	0.9	-27.7	-27.6	1.2	-22.7	31.8	20.7	27.5	20.9	2.7	340	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	85.1	67.0	81.9	65.5	78.8	64.4	69.6	80.3	67.7	77.9	66.0	75.7	6.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	100.0	74.4	64.1	93.0	72.5	62.3	87.3	71.1	34.4	80.2	32.8	77.9	31.4	75.8	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.7	21.3	18.4	-31.0	91.3	8.5	3.2	-37.1	93.6	-42.1	95.4	-46.9	97.2	-53.1	99.5		
			-31.5	73.3	8.2	2.4	-37.4	75.0	-42.2	76.4	-46.8	77.8	-52.7	79.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	2.3	9.1	23.1	39.0	52.5	62.6	66.2	61.8	50.5	38.2	21.6	8.5	(6)
(7)		DBStd	24.28	14.62	12.80	11.06	10.01	9.26	7.21	5.64	6.53	7.66	9.47	12.42	13.89	(7)
(8)		HDD50	6502	1478	1144	833	348	85	6	0	2	87	379	853	1287	(8)
(9)		HDD65	10624	1943	1564	1298	779	396	125	55	138	439	832	1303	1752	(9)
(10)		CDD50	1549	0	0	0	19	163	383	502	370	100	12	0	0	(10)
(11)		CDD65	197	0	0	0	0	9	53	92	40	3	0	0	0	(11)
(12)		CDH74	1997	0	0	0	4	190	550	785	405	63	0	0	0	(12)
(13)		CDH80	479	0	0	0	0	42	124	195	106	12	0	0	0	(13)
(14)	Wind	WSAvg	7.2	6.6	7.1	8.0	8.4	8.2	6.3	5.4	5.6	6.3	7.8	8.2	7.9	(14)
(15)	Precipitation	PrecAvg	28.1	1.7	1.3	1.7	1.8	2.3	3.0	3.3	2.9	2.3	2.3	3.1	2.6	(15)
(16)		PrecMax	35.2	2.9	2.6	4.5	4.1	5.2	5.5	5.0	4.3	5.2	5.1	5.3	4.9	(16)
(17)		PrecMin	22.4	0.3	0.2	0.4	0.5	1.1	1.6	1.5	1.3	0.8	1.2	1.3	0.9	(17)
(18)		PrecStd	3.1	0.7	0.6	0.9	0.9	1.0	0.9	0.9	0.9	0.9	0.9	0.9	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	36.9	51.7	71.9	85.3	87.7	88.6	89.5	81.0	68.2	50.0	36.7	(19)
(20)		MCWB	33.4	32.4	41.8	52.5	63.8	65.6	68.4	69.7	61.7	52.6	43.1	33.3	(20)	
(21)		2%	DB	30.8	32.8	45.0	66.3	79.1	83.5	84.8	82.6	73.7	61.1	45.2	32.4	(21)
(22)		MCWB	29.4	29.8	37.6	50.8	60.0	65.0	68.0	66.4	58.9	49.6	40.6	31.0	(22)	
(23)		5%	DB	26.4	29.9	40.9	60.9	75.0	80.3	81.8	78.6	68.4	56.8	39.6	28.9	(23)
(24)		MCWB	24.8	27.5	35.5	48.3	57.8	64.2	66.9	65.0	56.8	47.8	36.5	27.2	(24)	
(25)		10%	DB	22.2	26.6	37.4	55.5	70.1	76.8	78.6	74.9	64.0	51.7	35.9	26.4	(25)
(26)		MCWB	20.9	24.5	33.3	45.3	55.3	62.8	66.1	63.0	54.2	44.8	33.5	25.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.3	33.7	42.4	54.8	65.9	69.7	73.0	71.7	64.8	54.0	44.9	33.7	(27)
(28)		MDCB	34.6	35.4	50.3	67.8	81.2	79.9	83.1	84.1	76.2	64.5	48.9	35.2	(28)	
(29)		2%	WB	29.3	30.3	38.3	51.9	62.1	67.5	70.4	68.4	60.3	50.6	40.4	31.2	(29)
(30)		MDCB	30.6	32.3	43.5	62.5	75.4	79.1	80.5	79.8	71.3	60.0	45.1	32.4	(30)	
(31)		5%	WB	24.7	27.4	35.9	49.3	59.1	65.6	68.6	66.2	57.6	48.2	36.9	27.6	(31)
(32)		MDCB	26.2	29.7	40.3	59.7	71.3	77.1	78.1	76.1	66.6	55.7	39.8	28.7	(32)	
(33)		10%	WB	20.7	24.5	33.4	46.3	56.5	63.9	66.9	64.1	55.2	45.4	33.7	24.8	(33)
(34)		MDCB	22.1	26.5	36.9	54.5	67.4	74.1	75.7	73.1	62.2	51.1	35.6	26.4	(34)	
(35)	Mean Daily Temperature Range	MDBR	13.7	16.1	16.7	18.8	21.4	19.9	18.9	19.3	19.1	15.4	11.9	12.9	(35)	
(36)		5% DB	MCDBR	14.3	15.5	19.2	28.1	29.5	26.1	24.4	26.1	28.5	24.4	13.9	14.5	(36)
(37)		MCWBR	13.6	13.5	13.4	15.6	15.1	12.1	10.8	12.0	14.4	14.1	10.6	13.5	(37)	
(38)		5% WB	MCDBR	14.1	15.6	17.4	25.8	26.0	22.5	21.2	23.1	25.2	23.0	13.5	14.4	(38)
(39)		MCWBR	13.7	14.0	12.7	15.6	14.5	11.8	11.2	11.7	13.6	14.0	10.7	13.9	(39)	
(40)		Clear-Sky Solar Irradiance	taub	0.297	0.336	0.386	0.420	0.417	0.413	0.430	0.413	0.376	0.365	0.344	0.309	(40)
(41)	taud		1.998	1.937	1.873	2.023	2.176	2.272	2.237	2.263	2.308	2.201	1.997	1.936	(41)	
(42)	Ebn at Noon		194	226	241	253	263	266	258	255	250	218	170	156	(42)	
(43)	Edh at Noon		24	36	49	48	43	40	41	37	31	28	24	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	314	639	1058	1374	1643	1893	1743	1450	995	515	255	198	(44)	
(45)		RadStd	35	54	115	100	202	148	114	110	115	59	43	32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	-6.98	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page